

Lab 10 - Prompt and Specification Pack

Motion, Tracking and Temporal Continuity Measurement Generative AI for Image and Video Creation
(TGS-2020505925) | Version v11.0 | 6 September 2026

Motion work produces the acceptance evidence for a delivered clip. This pack is the continuity acceptance gate and the review wording that keeps 'it feels jittery' out of a defect report.

Currency note. Model identifiers and API parameters quoted anywhere in this pack were verified against the official vendor documentation on 6 September 2026. Model IDs change; re-verify before relying on one. Identifiers known to be deprecated are marked as such rather than quietly removed, so you can recognise them in older material.

Temporal continuity acceptance gate

```
{
  "gate": "clip_continuity_accept",
  "applies_to": "any delivered clip, generated or captured",
  "metrics": [
    {"name": "centroid_jitter_px", "definition": "std dev of frame-to-frame subject centroid step, excluding declared cuts", "max": 3.5},
    {"name": "scale_drift", "definition": "|tracker area ratio / expected area ratio - 1| over the clip", "max": 0.50},
    {"name": "mean_interframe_ssim", "definition": "mean SSIM between consecutive frames", "min": 0.82},
    {"name": "discontinuity_count", "definition": "frames where mean flow magnitude exceeds max(4x whole-clip median, 0.1 px); review candidates", "max_undeclared": 0}
  ],
  "note": "a declared cut is not a defect; an undeclared one is",
  "on_fail": "return with the failing metric, its value and the frame index"
}
```

Defect report wording (replace 'feels jittery')

```
CLIP: <file>   DURATION: <s>   FPS: <n>
DEFECT: centroid jitter 6.4 px (limit 3.5) measured on the subject track,
        concentrated between frames 88 and 112.
DEFECT: undeclared discontinuity at frame 100 - mean optical-flow magnitude
        11.2 px against a whole-clip median of 1.8 px.
NOT A DEFECT: scale change from frame 0 to 199 (expected, per shot list).
EVIDENCE: out/tracking.csv, out/continuity-report.json, out/track_camshift.mp4
```

Tracker selection note

mean shift	fixed window size; use only when the subject's apparent size is constant. Cheapest of the three.
CAMShift	recomputes window size and orientation from the back-projection each iteration; use when the subject approaches or recedes.
TrackerMIL	learns an appearance model online; benchmark partial occlusion robustness, slower, and built into the base opencv-python wheel.
CSRT / KCF	alternative trackers; benchmark quality. They are in opencv-contrib-python, NOT the base wheel.

Specify the dependency explicitly if you require them.

Continuity brief for a generated sequence

For each generated shot, deliver alongside the clip:

1. the subject track (per-frame centre and box) as CSV;
2. centroid jitter, scale drift and mean inter-frame SSIM;
3. a list of every frame index where a cut or transition is INTENDED.

Any discontinuity not on that list is a defect.

Do not smooth the track before reporting it - report the measured series.

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